Which process faults can be told apart? Diagnosability limits of in-situ monitoring in laser powder bed fusion

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Abstract

In-situ monitoring is standard on metal additive manufacturing systems, and classifiers report high accuracies for detecting defects from photodiode and imaging signals. Those results answer a question about outcomes: which defect formed. Specifying hardware requires an answer one step upstream — which process *causes* a configuration can separate in principle, and which remain confounded whatever classifier is applied. Where two causes produce signatures differing by less than the sensor noise, no volume of training data distinguishes them: the limit is structural, not algorithmic. This paper applies structural and quantitative diagnosability analysis, mature in automotive and aerospace process control, to laser powder bed fusion. A calibrated reduced-order thermal model gives the response of a coaxial suite to five injected faults — laser power and scan speed deviation, thin powder layers, degraded shielding gas flow and geometry-driven heat accumulation — with separability measured as an angle between noise-whitened signatures and each sensor's relative noise derived from its transduction chain rather than assumed. Laser power and gas degradation prove *exactly* confounded instantaneously, by algebra rather than noise, and seven candidate observables leave the attained rank at three against a structural bound of four. Regressing the same three sensors onto known build context raises the rank to five and the worst pair from 0.00° to 24.90° at no hardware cost, outperforming two idealised direct measurements of the causes themselves. Matrices are reported for five configuration classes found on current commercial machines.

Keywords: Additive manufacturing, Process monitoring, Diagnosability, Fault diagnosis, Laser powder bed fusion

1. Introduction

Laser powder bed fusion (LPBF) builds a part by melting thousands of short track segments per layer, and a modest excursion in any of several independent inputs can produce a rejected part. Manufacturers have responded by fitting in-situ monitoring as standard — coaxial photodiodes, melt pool cameras, pyrometers, increasingly acoustic and powder-bed imaging — and a large literature maps those signals onto defects with high reported accuracy.

Almost all of that work answers a question about *outcomes*: given the signals, which defect formed? That matters for part acceptance, but not for the machine builder specifying hardware or the process engineer facing a drifting build. Those concern *causes* — the laser is under-delivering, the recoater short-feeding, the gas flow degraded — and which it is determines what gets corrected. The distinction is not semantic, because causes can be mutually indistinguishable in a way outcomes are not: if two of them drive the readings along the same direction in signal space, the readings carry no information separating them and no classifier, however trained, recovers the difference. That limit is structural, a property of the measurement and the physics rather than of the algorithm, where a pair separable in principle but misclassified in practice is an algorithm that can be improved.

Diagnosability analysis is the established discipline for deciding which case holds. Developed within model-based fault detection and isolation, it asks, before any estimator is built, which faults a measurement set can detect and which it can isolate from one another [1]. The methods are standard where sensor sets are costly and must be justified in advance, but to the authors' knowledge have not been applied to LPBF.

This paper applies them. A calibrated reduced-order thermal model is driven by five injected fault modes and the signatures compared pairwise against noise, yielding a diagnosability matrix per configuration that divides fault pairs into those resolvable from existing signals, those needing an added measurement and those structurally irreducible. The contributions are the first such matrix for the process, to the authors' knowledge; noise figures derived from each sensor's transduction chain rather than assumed; a demonstration that laser power and gas degradation are *exactly* confounded instantaneously; the finding that the binding limit is the melt pool rather than the instrumentation; and that build structure already present dissolves it at no hardware cost.

2. Background and related work

2.1. What in-situ monitoring classifies

Melt pool monitoring with coaxial photodiodes and cameras is the most widely deployed configuration, and its canonical application is flaw classification. Scime and Beuth identified sig-

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natures indicative of keyholing porosity and balling from high-speed visible imaging [2], and later work extended this across modalities and alloys, including mechanical-property prediction from coaxial signals [3]. The predicted quantity is in each case an outcome or a material state, and whether two distinct excursions could have produced the same signal is never posed.

A smaller body of work estimates process inputs and is closer to the present question. RAISE-LPBF reconstructs laser power and scan speed independently from on-axis video on 316L, sampling both per scan line from continuous distributions so interactions can be studied [4]. Reijonen reports coaxial photodiode response to power, speed, focal position, scan orientation and shielding gas flow, finding power approximately linear, speed a power law and the remainder non-linear and interactive [5]. Both characterise *sensitivity*; identifiability — whether sensitivities to different causes are linearly distinguishable — is what neither addresses.

A third class observes the powder bed, attacking causes the melt pool reports only indirectly [7, 8], and returns in Section 4.7 as a target rather than a modelled option.

2.2. Lumped process metrics are incomplete

A recurring pattern is to report a *lumped* process quantity in place of its components. Volumetric energy density (VED) is the usual choice, and state-of-the-art parameter estimation from melt pool imagery still predicts energy density *class* rather than power and speed separately [9].

There is direct evidence that this discards information: Guo et al. held input energy density constant while varying power and speed together and saw melt pool dimensions change by orders of magnitude with transitions between melting regimes [10], and the 2022 Additive Manufacturing Benchmark found single-track depth correlated with VED while width did not [11]. Both are reproduced in Section 4.1. Falling back on a lumped group is the natural response to confounded components, and whether they are confounded for a given suite is the diagnosability question the literature has not asked.

2.3. Diagnosability analysis

Model-based diagnosability analysis operates on a model expressed over known variables, unknown variables and faults. Structural analysis considers only which variables appear in which equations, determining detectability and isolability from the resulting bipartite graph, with minimal structurally overdetermined (MSO) sets identifying the redundant subsystems from which consistency tests are built [1]. Being purely combinatorial it is insensitive to parameter error but can be optimistic: two faults may be structurally isolable yet numerically indistinguishable. Quantitative analysis supplies the remainder by linearising the fault-to-measurement map and examining the geometry of the signature directions; this paper runs both.

3. Method

3.1. Forward model and calibration

Melt pool geometry is computed from the Eagar–Tsai solution for a moving Gaussian heat source on a semi-infinite solid, extended with the Gladush–Smurov keyhole depth correction following Honarmandi et al. [12]. Four parameters — effective absorptivity, effective conductivity, heat capacity, the keyhole constant — are fitted to the seven IN718 bare-plate single-track cases of the 2022 Additive Manufacturing Benchmark [11], giving **11.7 % mean error on melt pool width and 13.1 % on depth**. All results below are evaluated at that benchmark's own baseline condition, 285 W, 960 mm s^{-1} and a 67 µm D4σ spot, where the errors are −15.5 % on width and +1.7 % on depth against expanded uncertainties ($k = 2$) of 6.2 and 14.1 µm respectively. Width is under-predicted at all seven conditions, the one-sided bias of a conduction solution with no mechanism that widens a pool. The keyhole correction is necessary rather than optional: all seven cases keyhole, depth-to-width ratios running 0.72–2.14 against the criterion's 0.67, and a pure conduction fit given the same three free parameters leaves 23.6 % depth error against 13.1 %, at an essentially unchanged 8.5 % on width. The quadrature in the history integral and the derivative step are both set by convergence tests: central differences are evaluated at $h = 0.002$, entries still moving 39.3 % between $h = 0.04$ and $h = 0.02$ but below 0.3 % at $h = 0.005$, the residual following the predicted second-order scaling.

3.2. Fault definitions

Five causes are modelled, each defined as a *physical* quantity acting on a model input so no arbitrary severity-to-signal constant is introduced. For magnitude f: *power* scales laser power, $P(1 + f)$; *speed* scales scan speed, $v(1 + f)$; *gas* is a plume optical depth attenuating effective absorptivity, $A\,e^{-f}$; *geometry* is a preheat rise in units of 100 K, $T_0 + 100f$; *layer* is a fractional layer-thickness deviation, positive denoting a *thinner* layer as produced by recoater short-feed, acting through three channels.

Two of those three channels derive from published measurement. The absorptivity coupling, −0.230, follows from the powder-to-solid absorptivity ratio of approximately two measured calorimetrically at 1070 nm [13] with a radiative penetration depth of a few particle diameters, implying a bare-solid absorptivity of 0.258, inside the 0.25–0.35 reported for nickel alloys near 1 µm — a consistency the calibration was not built to produce. The powder-melting enthalpy coupling, +0.186, follows from an energy balance in which melting the layer consumes 15.7 % of the absorbed energy, so a thinner layer frees proportionally more for penetration. The conductivity coupling remains an unmeasured estimate, swept across its range in Section 4.5 with every accumulated-regime result invariant to it, but it is load-bearing instantaneously: the other two channels nearly cancel on the absorbed-power product ($-0.230 + 0.186 = -0.044$), so without it the layer fault acts only through that product and joins the power/gas degeneracy *exactly* — its Jacobian column becomes −0.044 times the power column on all three sensors, and power/layer and layer/gas both read 0.00°. This one unmeasured coupling is all that distinguishes a thin layer from a weak laser at an instant. All couplings are applied in exponential form $A\,e^{cf}$, agreeing with $A(1 + cf)$ to first order but unable to pass through zero, where the linear form would drive thermal diffusivity negative and the integrand imaginary.

3.3. Sensor models

Three coaxial observables are computed from the temperature field: a band-integrated photodiode (700–1000 nm), melt pool area, and a two-colour pyrometer (750/950 nm) evaluated as the ratio of integrated band powers and inverted for temperature. Spectral radiance follows Planck's law, temperature being capped at the boiling point before any radiance is computed. Four candidates are defined for Section 4.3: the camera's width and length reported separately, a short-wave infrared photodiode and a wide-band pyrometer.

3.4. Diagnosability measures

The relative-sensitivity Jacobian is

$$J_{sk} \;=\; \frac{1}{y_s}\,\frac{\partial y_s}{\partial f_k}, \tag{1}$$

so entry (s,k) is the fractional change in sensor s per unit of fault k. Each row is then divided by that sensor's relative noise σ_s, derived in Section 3.6, to give the whitened matrix J^{w}: only then are rows in a common unit and angles between columns physically meaningful.

Three quantities are reported throughout. The *angle* between whitened columns, $\theta_{kl} = \arccos\left(|\langle J^{\mathrm{w}}_{:k}, J^{\mathrm{w}}_{:l}\rangle|/\|J^{\mathrm{w}}_{:k}\|\|J^{\mathrm{w}}_{:l}\|\right)$, measures pairwise separability: 0° means the two faults move the readings along one line and are indistinguishable by any estimator, 90° that they are orthogonal. A practical floor of 10° is adopted as a convention and its influence reported. The *rank* of J^{w} counts how many fault directions are observable at all, treating a singular value below 10^{-6} of the largest as none, and the *condition number* how thin the weakest retained one is; rank without it is misleading, as Section 4.3 shows. Structural analysis is run in parallel on an eleven-equation representation using MSO sets, as an independent check.

3.5. Build context and the enlarged observation

A build is not a single instant. It comprises thousands of segments whose toolpath, layer index, gas-flow direction and local geometry are known to the machine, and the five faults do not vary identically across them: laser degradation drifts with time, galvanometer error concentrates at direction reversals, a recoater short-feed recurs with layer index, gas degradation follows the flow direction, heat accumulation follows support depth. Treat the build rather than the instant as the observation: let $G \in \mathbb{R}^{N\times 5}$ hold five known context regressors over N segments — a constant, position along the gas-flow axis, layer parity, turn density, support depth — and $M \in \mathbb{R}^{N\times 5}$ the modulation profile of each fault. Regressing each sensor's relative residual onto G gives five coefficients per sensor in place of one reading, so three sensors give 15 observables, with Jacobian

$$J^{\mathrm{big}}_{(s,j),k} = J_{sk}\,B_{jk}, \qquad B = (G^{\top}G)^{-1}G^{\top}M. \tag{2}$$

The noise on the j-th coefficient from sensor s is not σ_s but $\sigma_{(s,j)} = \sigma_s\sqrt{[(G^{\top}G)^{-1}]_{jj}}$, the square-root factor running 0.109 to 0.136, so fitting $N = 84$ segments to five regressors buys a seven- to ninefold noise reduction. Results use $N = 84$, with the dependence on N in Section 4.5.

The profiles in M are physical claims rather than measurements, and they are *machine-specific*: a different nozzle geometry, scan strategy or part gives different patterns, so there is no universal set of five to go and find. One of them can nonetheless be calibrated from published data, and the inversion is the recipe by which a reader calibrates the rest on their own machine. The AM Bench pad scans [11], under a 4.3 m s^{-1} cross-flow, show melt pool depth falling from 175 to 125 µm on tracks scanned parallel to the gas flow against anti-parallel ones — a 27.2 % mean reduction against standard deviations of 4–11 µm — with perpendicular pads as a control at 2.4–4.7 %. Weaver et al. attribute the reduction to plume by-products attenuating, scattering and lensing the beam, and report that none of the four challenge submissions predicted it; this model represents the attenuation alone. Inverting each depth change through the model's gas channel at the benchmark's baseline condition — where the model's depth error is +1.7 % against −15.5 % on width — gives $f_{\mathrm{gas}} = 0.046$ for the control against 0.404 for the parallel mean, an **8.8× swing within one healthy build** against the 6.05× assumed here. In contrast units that is 1.08 where this work assumes 0.90, so **the one profile that can be checked is understated by 20 %**. Three qualifications: the driver is scan direction rather than position along the flow, it is baseline plume behaviour rather than a fault, and the inversion is model-mediated. The gas signature also alternates at track scale where a recoater signature alternates at layer scale — the distinctness Section 4.5 shows the method needs.

3.6. Sensor noise, derived from transduction

Whitening is half of the measurement geometry, so an asserted noise set silently decides the answer. Only one in-process figure proved citable, a 2.3 % relative width uncertainty from a cross-sectional metrology budget [11], which is not in-process noise. The figures are therefore *derived* from one assumption — each radiometric channel carries the same fractional noise ε — with the rest following from how each instrument turns radiance into its reading. Only ratios enter the whitening, so ε cancels and no absolute threshold is claimed.

The photodiode reading *is* a band-integrated radiometric power, so $\sigma_{\mathrm{pd}} = \varepsilon$ and it serves as the unit. The pyrometer inverts a ratio of two band powers: independent channel noise puts $\sqrt{2}\,\varepsilon$ on the ratio, and the inversion divides by the ratio's Wien temperature sensitivity $|\partial \ln R/\partial \ln T| = (c_2/T)(\lambda_2^{-1} - \lambda_1^{-1}) = 1.333$ at 750/950 nm and the model's 3031 K, giving $\sigma_{\mathrm{pyr}} = 1.061\,\varepsilon$. Closely spaced bands discard most of the available temperature sensitivity and the inversion amplifies what noise remains, so a two-colour pyrometer is a *noisier* relative instrument than the photodiode it is built from. The camera reports an extent found by thresholding at the liquidus, where a fractional radiance error is indistinguishable from the threshold sitting at a different temperature, $\delta \ln T_{\mathrm{liq}} = -\varepsilon/(\partial \ln L/\partial \ln T)$. Planck is steep there, $\partial \ln L/\partial \ln T = 10.52$ at 850 nm and 1609 K, so $\varepsilon = 1$ % displaces the apparent isotherm by only 0.095 % in temperature, and propagating that shift through the model's own dimension solver gives $\sigma_{\mathrm{cam}} = 0.142\,\varepsilon$: thresholding on a steep Planck function localises an edge well, making the camera the most precise of the three by a factor of seven.

The derived set is 1 : 0.142 : 1.061, against the 1 : 1 : 0.03 assumed before deriving it — wrong in both free entries and in opposite directions. Two choices inside the derivation were taken in the direction that costs this analysis: the area channel is not averaged along the pool perimeter, and width and length are whitened as independent although a common-mode error correlates them, which understates the value of reporting them separately. Both limits are swept, alongside a gain-dominated case in which the pyrometer's ratio rejects common-mode drift and approaches the 0.03 once assumed.

4. Results

4.1. Validation

One external check and one of internal structure. Against the constant-energy-density experiment of Guo et al. [10] the model achieves a **partial pass**, reproducing the existence, direction and regime transition of the variation but spanning 2.37× in depth where the experiment reports orders of magnitude — the mechanism they identify, absorption varying with vapour depression geometry, being absent from a conduction solution. The same sweep yields a result used in Section 4.7: at constant energy density melt pool *length* varies by 2.53× against 1.11× for width and depth.

The second asks whether the model reproduces the benchmark's reported structure, depth tracking volumetric energy density while width does not [11]. As a correlation it is unreliable — over 27 melted conditions the model gives $r = +0.20$ for width against +0.77 for depth, but width's coefficient runs +0.68 to −0.22 across five such grids, a property of the sampled region rather than of the model. The exponents are stable where the correlation is not. Width and depth are each a power law in (P, v, d) to $R^2 \geq 0.94$, and a pure function of VED $= P/(vd^2)$ must have exponents along $(1, -1, -2)$: on every grid depth's lie at $\cos 0.64$–0.74 of that ray against width's −0.11 to 0.00, because the two depend on spot size with opposite signs — the one variable a volumetric density folds away. All 27 points are model output, so this is a consistency check, not external validation.

4.2. Laser power and shielding gas are exactly confounded

Both faults act *only* on the product (effective absorptivity × power), in which the Eagar–Tsai temperature rise is linear. Gas attenuates absorptivity as $A\,e^{-f_{\text{gas}}}$ and power scales as $P(1 + f_{\text{power}})$, so the product is unchanged whenever $f_{\text{gas}} = \ln(1 + f_{\text{power}})$, giving an identical melt pool — the same temperature at every point, hence identical readings from any instrument that could be built — the two faults cancel bit-identically in double precision on all three. The ratio of the two Jacobian columns is −0.99999, −1.00000, −0.99993 across the sensors, a spread of 7×10^{-5} from finite-difference error on an exact identity, and the pairwise angle is **0.00**°. The degeneracy is algebraic, not numerical: sweeping every coupling constant across its plausible range leaves it at 0.00°, and the structural analysis identifies the same pair from equation form alone.

4.3. The limit is the melt pool, not the sensor count

The three observables are not independent. Normalising the Jacobian's sensor Gram matrix JJ^{T} gives pairwise correlations of **0.99960** (photodiode–camera), **0.99508** (photodiode–pyrometer) and **0.99477** (camera–pyrometer): a photodiode reading band radiance, a camera reading pool area in square metres and a two-colour pyrometer reading kelvin respond to the five faults in patterns agreeing to better than 0.994. Row scalings cancel in the normalisation, so this is the one result here that does *not* depend on the derived noise figures.

The singular values quantify those correlations rather than establishing them independently — 13.14, 0.0638 and 0.00513, so 99.998 % of the suite's squared response lies along a single direction. Three of the five fault directions are observed and **two are not observed at all**, the unobserved pair spanning the power–gas combination of Section 4.2; of the three observed, the weakest responds **2562** times less than the strongest. Adding all four candidate observables gives seven rows and does not raise the attained rank: the singular values become $(23.7,\ 9.74,\ 6.0 \times 10^{-2},\ 5.0 \times 10^{-6},\ 1.4 \times 10^{-6})$, the fourth at 2×10^{-7} of the leading one, two orders below the 10^{-6} threshold adopted here, so four directions are formally present but only three are measurements. **Seven instruments leave power and gas exactly as confounded as one.**

The reason is structural. Everything the model knows funnels into four physical channels — absorbed power, scan speed, effective conductivity and preheat — with three of the five faults crowding onto the first. Any instrument reading only the melt pool is a function of those four channels, so the fault-to-sensor map cannot exceed rank 4 however many are fitted, and at realistic noise it does not reach even that: a new wavelength reweights the same Planck curve over the same pool, adding a row, not a direction.

4.4. Build context dissolves the limit

With the enlarged observation of Section 3.5, rank rises from 3 to **5**, the worst pair from 0.00° to **24.90**°, power versus gas to 29.34°, and the condition number falls from 2562 to 50 — all ten pairs clearing the 10° floor (Table 1) against none per segment.

Three controls establish that this arises from the claimed mechanism rather than the arithmetic. Most important, giving all five faults one identical profile returns the instantaneous result *exactly* — rank 3, condition number 2562, worst pair 0.00° — despite the enlarged matrix still having 15 rows, because proportional columns of B make it inherit the instantaneous column directions: **the gain is not dimensionality, it is the profiles differing**. Replacing every derivative by a secant through the full nonlinear chain gives rank 5 at every magnitude from 0.2 % to 20 % in both signs, worst pair 12.82°–37.95° and condition number 47–62, so the derivative limit is *not* strictly conservative: a +20 % fault gives 12.82°, above the floor by only 1.3×. Recomputing with the full model at every segment reproduces the angles to 0.04°.

4.5. Robustness

Because the profiles are machine-specific, the primary result is separability *as a function of* profile strength: five of ten pairs clear the floor at or below 25 % of the strength assumed here and the hardest, power/speed, at **45 %**, so a machine whose patterns are half as strong still resolves every pair — and the one profile calibrated against measurement came back 20 % *stronger* than assumed. The headline moves by under half a degree across the full range of the unmeasured conductivity coupling and all four noise sets of Section 3.6. Reassigning which fault follows which build variable moves it further, the worst pair running 16.2° to 30.1° across all 120 permutations, but **every one of the 120 leaves rank five and all ten pairs above the floor** — the result needs the patterns to differ, not any particular fault to own any particular one. Segment count is the one real requirement, at 11.23° for $N = 12$ and 22.90° for $N = 30$ on a near-square plan, flat thereafter.

What the method requires is not accurate contrasts but *distinct* patterns, and at these noise figures strictly so: forcing any two equal collapses that pair to its instantaneous angle, and every instantaneous angle is below the floor, the mildest leaving 4.63°. **No pair of faults may share a spatial signature.** None does on this machine — the recoater advances once per layer along a fixed axis while the gas flows along a fixed chamber axis, and the benchmark measurement of Section 3.5 confirms they alternate at different scales — and establishing that on another machine is a far weaker requirement than measuring five contrasts.

Distinctness also tolerates a profile being attributed to the wrong build variable, which matters because measured layer-thickness variation is reported to follow part support rather than the recoater cycle assumed here [7]. Mixing the geometry fault's support driver into the layer profile leaves layer/geometry above the floor until that profile is roughly 84 % support-driven (12.07° at 80 %), and mixing the power fault's build-drift driver never brings power/layer below it at any fraction, since every profile is normalised to unit mean and two faults sharing a driver still differ wherever their contrasts do.

Table 1: The diagnosability matrix, in degrees; bold clears the 10° floor. Power/gas reads 0.00° in both melt-pool columns, being algebraic. The free split clears seven pairs and mildly worsens two that stay below the floor either way; accumulation over $N = 84$ clears all ten.

	per segment		over a build
fault pair	baseline	split camera	baseline
power / gas	0.00	0.00	**29.34**
layer / gas	0.32	**64.53**	**44.48**
power / layer	0.32	**64.53**	**36.26**
power / speed	0.70	**43.43**	**24.90**
speed / gas	0.70	**43.43**	**35.87**
speed / layer	1.03	**72.04**	**46.16**
speed / geometry	3.62	**43.28**	**31.95**
power / geometry	4.31	3.08	**25.42**
gas / geometry	4.31	3.08	**31.39**
layer / geometry	4.63	**64.81**	**40.54**
pairs clearing the floor	0/10	**7/10**	**10/10**

4.6. Diagnosability matrix and configuration classes

Classifying on the enlarged observation alone places all ten pairs in class 1 for every configuration, a single photodiode included — true but uninformative. Two regimes are therefore reported (Table 1): *instantaneous*, per-segment decisions for closed-loop control, and *accumulated*, diagnosis over a build for qualification and drift monitoring. The baseline suite resolves none of the ten instantaneously and all ten accumulated, class 3 empty in both.

Five commercial systems were surveyed and present only three distinct observable sets. Those are reported as classes rather than products, alongside a single-photodiode floor and the research reference: **A**, a single coaxial photodiode (rank 1); **B**, photodiode plus pool imaging, and **C**, two photodiodes in separate bands (rank 2); **D**, the research reference adding pyrometry, and **E**, C plus upstream power monitoring (rank 3). No vendor is named: mapping a product to a set of observables rests on secondary literature rather than datasheets, detection bands being withheld and published descriptions of one product disagreeing.

No configuration built solely from melt-pool observables separates laser power from gas degradation, whatever its sensor count: A through D all read 0.00° on that pair, for the reason given in Section 4.3, and at the derived noise figures none resolves *any* pair per segment. E is the actionable class — a measurement of delivered laser power sampled *upstream* of the process zone separates the pair at 84.02° and resolves four of ten, seeing the laser output without the plume attenuation. One surveyed system may belong at E rather than C depending on whether its laser-energy photodiode samples upstream, and is classed by what public documentation states rather than by what is likely. B and C are separated not by class count, both empty, but by correlation: two bands of the same emitted light agree at 0.99996 against 0.99960 for a geometric measurement.

4.7. What an added measurement would buy

The sensors most likely to help — plume sensing for gas, powder-bed or recoater sensing for layer [7, 8] — cannot be simulated by a model with neither a plume nor a powder layer, but what such a sensor *measures* can be: a unit row on fault k at a given relative noise, the idealised best case for any instrument aimed at that cause.

A direct measurement of any one cause raises the rank to 4 and clears exactly the four pairs involving that cause — no more. Geometry is the exception, raising the rank not at all: a unit row on geometry already lies *entirely* within the row space of the existing three (projection 1.0000, against 0.53–0.91 for the others), so the suite observes preheat already and simply cannot separate the rest. **Precision is irrelevant**: sweeping the added sensor's noise from 50 % to 0.2 % leaves the worst pair at 0.32° and four of ten cleared throughout, since adding a row in a new direction changes the rank while making an existing row quieter does not. **Build context outperforms two idealised, perfect direct measurements of the causes themselves**: gas and layer sensors together give rank 5 but 0.70° and seven of ten, against 24.90° and ten of ten for build context at no cost — neither says anything about power versus speed.

Among melt-pool observables the most valuable addition is free, and is the strongest practical result here. Reporting the camera's two axes separately rather than as an area is worth **+23.89**° on the pair that binds the baseline accumulated result (24.90° to 48.79°), after which power/geometry binds instead at 25.26°, so the suite's worst accumulated pair gains only 0.35°. What the split buys outright is *per segment*: it is the only change resolving anything there, taking that count from none of ten to **seven of ten** (Table 1); a short-wave infrared photodiode (+0.01°) and a wide-band pyrometer (0.00°) are worth nothing. The mechanism is independence, not responsiveness: width is both the candidate least correlated with the coaxial photodiode, 0.779 against 0.979 for length, and the better localised, its edge on a steep lateral gradient where the trailing edge lies in the pool's thermal tail (Section 3.6). An area averages the informative axis against the uninformative one and buries it, while responsiveness runs the other way — length carries 2.5× the variation of width (Section 4.1) and still adds less.

Failed and successful candidates differ by one criterion: *a sensor added to break a confound must observe upstream of the shared channel; a sensor reading the melt pool adds a row, not a direction.*

5. Discussion

5.1. Relation to reported monitoring performance

Comparing predicted limits against reported accuracies proves largely unrealisable, and the reason is itself a result. Of 19 works surveyed — in-situ LPBF monitoring with a stated input signal and output, 2019–2025 — seven classify outcomes and only **three estimate a process input from an in-situ signal**, one of those a lumped energy-density class. None analyses whether two causes are separable, and the one work varying power, speed and gas flow on a single photodiode [5] does not frame the question that way. That is not an accident of what the field chose to study: outcomes are functions of the melt pool, and the melt pool is a function of the four channels of Section 4.3, so outcome classification cannot separate two causes sharing a channel, for the same reason no melt-pool instrument can. Absence of evidence is not evidence of absence, but no published result contradicts the predicted limits, and two groups independently report the lumped group this analysis identifies as what survives confounding [4, 9].

The one external check on this model's sensor response can be made, and it partly fails. Reijonen reports the coaxial photodiode response to scan speed as $y = ax^{-b}$ with $b = 0.337$ at 400 W over 400–1200 mm s^{-1} [5]. The comparable quantity is a power law fitted to this model over *their* range rather than the Jacobian entry −0.483; that fit gives **0.468**, some 39 % too steep. The qualitative comparison is worse: they find b growing with laser power, "almost non-existent" at 100 W and reaching 0.337 at 400 W, where this model gives 0.51, 0.44 and 0.47 at 100, 285 and 400 W — flat.

The discrepancy lies in physics this model assumes away. Reijonen attributes the speed dependence to the vapour plume: the trend holds only where gas flow is high and constant, and at low power, in conduction mode, there is no plume and no speed effect at all. Here the plume is a *fixed* optical depth, independent of power and speed and emitting nothing of its own, so no mechanism exists by which b could grow with power. That same assumption is what makes the power/gas degeneracy exact (Section 4.2), and these measurements show the real attenuation depends on both parameters and can even reverse sign with scan speed at low gas flow. **The reported** 0.00° **is an upper bound on the instantaneous confounding on external evidence, not merely by assertion.**

Three caveats on the comparison: it crosses material (316L against IN718), detection band (a shortpass below 800 nm against 700–1000 nm) and bed condition (30 µm powder layers against bare plate). One point in the model's defence: it is calibrated at 285 W where all seven benchmark cases keyhole, so the fixed optical depth is a local approximation *inside* the regime where a plume exists rather than a claim about conduction-mode melting. And on the power response, Reijonen reports a linear fit $y = ax + b$ without quoting the intercept; since relative sensitivity is then $aP/(aP + b)$, equal to one only through the origin, this model's +1.52 cannot be called a conflict until that intercept is known.

5.2. Limitations

The thermal model is a conduction solution with an analytic keyhole patch: it partial-passes the Guo validation, its fitted parameters sit at their bounds, and the absorptivity, conductivity and heat capacity correlations of 0.93–0.96 reported during Bayesian calibration of the same model family [12] are the pathology this paper studies appearing in its own calibration. Three modelling choices then make the instantaneous case *more* confounded than reality: the gas fault as pure optical depth omits a convective channel; treating plume attenuation as independent of power and speed is what makes the degeneracy exact; and modelling upstream power monitoring as a melt-pool-derived signal is likewise pessimistic.

The 0.00° **is therefore an upper bound, and its exactness is an artefact of the one-channel gas model.** Gas here acts only on absorbed power, precisely the channel a power fault acts on, which is what makes the two algebraically identical. A real plume also scatters: Reijonen et al. report reduced gas flow giving loss of penetration together with *increased* melt pool width [6], where attenuation alone gives a decrease — −10 % here at $f_{\mathrm{gas}} = 0.4$. Adding that broadening as a spot-size channel, which power does not possess, opens the pair to **2–3**° at the strength needed to reverse the width sign, and leaves the suite's worst pair unmoved at 0.32°. The exactness is a property of the model; the inseparability is not. The rank limit is unaffected either way, deriving from channel counting rather than the form of this coupling.

The geometry inversion fails, returning +687 K of preheat because a pad differs from a single track by remelting as well as preheat, and no dataset yet offers a build with *deliberately induced* machine faults: every inversion here reads a healthy build and extrapolates, which is the natural follow-on experiment. The noise derivation rests on one assumption, that every radiometric channel carries the same fractional noise, bounded

by the gain-dominated sweep. The five fault magnitudes are physical but not on a common severity scale, so comparisons *across pairs at fixed magnitude* are not meaningful, and all results are diagnosability statements — what a suite can distinguish in principle, at a given relative noise, under a given forward model — not that any algorithm achieves it.

6. Conclusions

Laser power deviation and shielding gas degradation produce identical melt pools in this model, confounded at 0.00° by algebra rather than noise, so no instrument reading the melt pool separates them from a single instant. The binding limit is the melt pool's four physical channels, not the instrumentation: at noise derived from each sensor's transduction chain, seven observables attain rank 3 against that bound of 4 and leave the worst pair at 0.00°. Deriving those figures rather than assuming them matters, whitening being half of the measurement geometry — the set assumed before deriving it was wrong in both free entries and in opposite directions.

Build structure already present dissolves the limit at no hardware cost — rank 3 → 5, worst pair 0.00° → 24.90°, all ten pairs resolved where none is per segment — surviving at half the assumed profile strength, under all four noise sets, at fault magnitudes from 0.2 % to 20 %, and in all 120 reassignments of profile to fault, with a shared-profile control reproducing the instantaneous result exactly. It outperforms two idealised direct measurements of the causes themselves and requires only that no two faults share a spatial signature; those patterns being machine-specific, the deliverable is the method plus the strength a machine needs, with the inversion of Section 3.5 as the recipe for calibrating it.

For real-time control an upstream measurement of delivered laser power is the recommendation, separating power from gas at 84° where no melt-pool configuration separates them at all; among melt-pool observables the most valuable addition is free, reporting the camera's two axes separately being worth +23.89° on the pair that binds the baseline and being the only change resolving anything per segment. Post-build diagnosis needs no new hardware, real-time isolation does, and crude sensing of the right cause suffices. The method needs only a forward model and a sensor description, and answers at specification time what is otherwise settled after the hardware is bought.

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